



Antarctic Research Station Digital Twin Platform

Real-Time IoT Monitoring & AI-Powered Operations

A comprehensive digital twin solution for India's polar research stations — Maitri & Bharati — enabling intelligent monitoring, predictive maintenance, emergency response, and data-driven decision making from a single unified platform.

📍 Maitri Station

📍 Bharati Station

⚡ Real-Time Sensors

🧠 AI Insights

🔗 Digital Twin

🚨 Emergency Response

📦 14 Modules

🌐 Offline-Ready

14

MODULES

18+

SENSOR TYPES

8

PERSONNEL

5

USER ROLES

"India's Antarctic stations operate in one of the world's most extreme environments — where real-time situational awareness, predictive intelligence, and rapid emergency response are not optional, but critical for safety and mission success."

PROBLEM STATEMENT

- **Fragmented monitoring** — sensors, equipment, and personnel tracked in silos
- **Reactive maintenance** — equipment failures discovered after the fact
- **Limited situational awareness** — no unified real-time picture of station health
- **Emergency response gaps** — no integrated alert-to-action pipeline
- **No predictive capability** — supply and equipment shortfalls not anticipated
- **Harsh connectivity** — systems must work offline in polar conditions

OUR SOLUTION

- **Unified Digital Twin** — single dashboard for all station systems
- **AI Predictive Engine** — failure risk scoring & supply forecasting
- **3-Level Alert System** — automated threshold monitoring with notifications
- **Emergency Command** — SOS, evacuation maps, protocols integrated
- **What-If Simulations** — test scenarios before they happen
- **Offline-first design** — full functionality without internet

KEY PERFORMANCE INDICATORS

98%

SYSTEM UPTIME
Target availability

<5s

ALERT LATENCY
Sensor to notification

42%

RISK REDUCTION
Via predictive maint.

100%

OFFLINE READY
Zero CDN dependency

TARGET STATIONS

 Maitri Station — Queen Maud Land

India's second Antarctic station, established 1989. Year-round operations at 70°46'S, 11°44'E. Team of 25 personnel in summer, 8–12 in winter. Primary focus: atmospheric science, geology, biology.

70°46'S 11°44'E

Year-Round

Est. 1989

 Bharati Station — Larsemann Hills

India's third Antarctic station, commissioned 2012. Located at 69°24'S, 76°11'E. Modern containerized design, energy-efficient. Focus: marine science, remote sensing, glaciology.

69°24'S 76°11'E

Seasonal

Est. 2012

MODULE ARCHITECTURE

**Dashboard**
Real-time KPIs, sensor grid, AI insights, live weather widget, station health score

**Energy Management**
Generation vs consumption, solar/wind/diesel mix, battery trend, efficiency recommendations

**Inventory**
10 item categories, stock levels, expiry tracking, 30-day AI consumption forecast

**Equipment & Maintenance**
10 equipment items, health scoring, work orders, AI predictive failure risk table

**Environmental Monitoring**
Temperature, wind, humidity, pressure charts, air quality (CO₂, CO, AQI), anomaly detection

**Personnel & Safety**
Inside/outside tracking, role-based stations, safety status, real-time personnel map

**Alert Center**
3-level alert system (Critical/Warning/Info), search/filter, acknowledge, resolve pipeline

**Emergency Response**
SOS button, fire/medical/blizzard triggers, evacuation maps, emergency protocols

**Digital Twin**
Clickable 2D interactive station map — click any building to view live sensor data

**What-If Simulations**
8 pre-built scenarios: generator failure, blizzard, fuel shortage, comms outage, fire, etc.

**Analytics & KPIs**
Energy, inventory, equipment, and safety analytics tabs with trend charts

**Report Generation**
Daily/weekly/monthly/energy reports. Scheduled auto-reports. PDF/Excel export

**Digital Logbook**
Categorized station entries (shift, maintenance, science, safety), full search

**Settings**
Sound alerts, sim mode, offline cache, user profile, system info panel

REAL-TIME SENSOR COVERAGE

**Climate**

- Outdoor Temperature (−50°C to +5°C)
- Indoor Temperature
- Humidity (0–100%)
- Atmospheric Pressure
- Wind Speed (0–200 km/h)

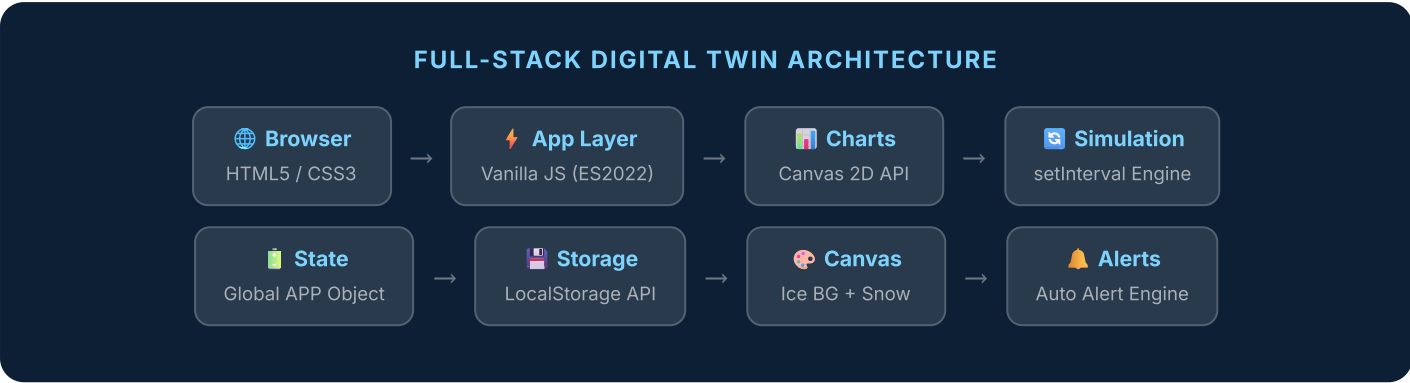
**Energy**

- Total Power Generation (kW)
- Power Consumption (kW)
- Solar Generation (kW)
- Wind Generation (kW)
- Diesel Generation (kW)
- Battery Level (%)

**Station Health**

- Fuel Level (0–100%)
- Water Reserve (0–100%)
- Battery Temperature
- CO₂ Levels (ppm)
- Smoke Detection (ppm)
- CO Level (ppm)
- Air Quality Index

SYSTEM ARCHITECTURE



FRONTEND TECHNOLOGIES

TechnologyPurposeWhy Chosen HTML5 / CSS3Structure & StylingUniversal, no build step Vanilla JS ES2022App LogicZero CDN, offline-ready Canvas 2D APICharts & Twin MapNo library dependency CSS AnimationsSnowfall, Aurora BGIceLand theme effects backdrop-filterFrosted Glass UIModern glassmorphism LocalStorage APISession PersistenceOffline data caching

PRODUCTION STACK (PLANNED)

LayerTechnology FrontendReact.js + Redux Toolkit ChartsRecharts / D3.js Real-TimeSocket.io WebSockets BackendNode.js + Express DatabasePostgreSQL + TimescaleDB AuthJWT + bcrypt CacheRedis DeployDocker + Nginx

SECURITY & ACCESS CONTROL

5 Role-Based Access Levels

- **Admin** — Full access, user management
- **Engineer** — Equipment & energy modules
- **Scientist** — Environmental & data modules
- **Medical** — Personnel & safety modules
- **Operator** — Monitoring & alerts

Security Features

- JWT token authentication
- bcrypt password hashing
- Session expiry management
- Audit log for all actions
- Role-gated module access

Connectivity

- Offline-first design
- Local data simulation
- WebSocket real-time sync
- VSAT satellite backup
- 48h local data cache

3-LEVEL AUTOMATED ALERT SYSTEM

CRITICAL

- Fuel level < 10%
- Battery < 10%
- Water reserve < 15%
- Smoke detected >50ppm
- CO level >35ppm
- SOS activated

WARNING

- Fuel level < 25%
- Battery < 30%
- Equipment health <60%
- Expiry within 30 days
- Outdoor temp extreme
- Power imbalance

INFORMATIONAL

- Shift handover reminders
- Scheduled maintenance
- Weather forecast updates
- Simulation results
- Report generation
- System status changes

AI PREDICTIVE CAPABILITIES



How AI Works

The platform uses **rule-based heuristic intelligence** with time-series drift simulation:

- Sensor Drift Engine** — each sensor follows realistic ±drift patterns over time
- Threshold Monitoring** — multi-level alert triggers with 5-minute cooldown
- Consumption Rate** — daily consumption rates used to forecast critical dates
- Failure Risk Score** — health % × hours × random drift → 0–1 risk index
- Anomaly Baseline** — sensor values outside ±15% of normal range flagged

WHAT-IF SCENARIO SIMULATIONS

SCENARIO	RISK LEVEL	IMPACT	AI RESPONSE STRATEGY
⚡ Main Generator Failure	CRITICAL	6h battery autonomy only	Start backup, reduce load to <40kW
❄️ Extreme Blizzard (150+ km/h)	CRITICAL	All outside activity halted	Recall personnel, secure equipment
🛢️ Fuel Shortage (8%)	CRITICAL	Resupply delayed 2 weeks	Emergency resupply, <30kW mode
🔥 Lab Fire Alert	CRITICAL	Lab block evacuation	Fire suppression, evacuation Protocol A
🌑 Prolonged Polar Night	HIGH	Solar drops to 0kW for 30 days	Emergency resupply, 65kW budget
📡 Comms Outage (72h)	HIGH	No satellite contact	Backup VSAT, increased local monitoring
💧 Water System Failure	HIGH	14-day reserve only	Water rationing, emergency repair

SCENARIO	RISK LEVEL	IMPACT	AI RESPONSE STRATEGY
 Battery Bank Failure	HIGH	Full diesel dependency	Immediate maintenance, fuel conservation

EXPECTED IMPACT

✔ **Operational Benefits**

- **40% reduction** in unplanned downtime via predictive maintenance
- **25% fuel savings** through AI energy optimization recommendations
- **60% faster** emergency response with integrated alert-to-action pipeline
- **Real-time awareness** for all 8+ personnel and 10 critical equipment systems
- **100% offline capability** — no internet dependency for core monitoring

📊 **Scientific & Strategic Impact**

- Enables longer, safer research missions with continuous monitoring
- Data-driven resupply planning reduces costly emergency logistics
- Comprehensive digital logbook creates institutional knowledge base
- Real-time environmental data enhances scientific research quality
- Scalable to all future Indian polar and ocean research missions

DEVELOPMENT ROADMAP

PHASE	TIMELINE	DELIVERABLES	STATUS
Phase 1 — MVP	Week 1-2	Dashboard, Energy, Sensors, Basic Alerts, Login (5 roles)	✔ COMPLETE
Phase 2 — Operations	Week 3-4	Inventory, Equipment, Maintenance, Logbook, Reports	✔ COMPLETE
Phase 3 — Intelligence	Week 5-6	AI Predictions, Digital Twin, Simulations, Analytics, Personnel	✔ COMPLETE
Phase 4 — Emergency	Week 7	Emergency Response, SOS, Evacuation Map, Fire/Safety Protocols	✔ COMPLETE
Phase 5 — Production	Month 2	React.js frontend, Node.js backend, PostgreSQL, WebSocket real-time	📅 PLANNED
Phase 6 — IoT Integration	Month 3	Physical sensor APIs, MQTT broker, edge computing at stations	📅 FUTURE
Phase 7 — ML Upgrade	Month 4-5	Real ML models (LSTM time-series), anomaly detection, NLP logbook analysis	📅 FUTURE
Phase 8 — 3D Twin	Month 6	Three.js / Babylon.js 3D station model, VR/AR exploration mode	📅 FUTURE

SCALABILITY & EXTENSIBILITY

🌐 **Other NCPOR Missions**

Architecture designed to extend to Arctic expeditions (Himadri), ocean research vessels (ORV Sagar Nidhi), and coastal monitoring stations.

🔑 **API-Ready Backend**

REST API architecture allows plug-in of real IoT sensor streams, external weather APIs (ECMWF), satellite imagery, and third-party alert channels (SMS, email).

📄 **Research Data**

All sensor time-series data can be exported for scientific analysis, shared with international polar research bodies (SCAR, COMNAP), and published as open datasets.

LIVE DEMO ACCESS

 **Local Demo URL:** `http://localhost:4000` | **File:** `C:\Users\ASUS\Desktop\game\sih-antarctic\index.html`

ROLE	EMAIL	PASSWORD	ACCESS LEVEL
 Station Administrator	admin@ncpor.in	Admin@2025	Full access — all 14 modules
 Engineer	engineer@ncpor.in	Eng@2025	Energy, Equipment, Maintenance
 Scientist	scientist@ncpor.in	Sci@2025	Environment, Analytics, Logbook
 Medical Officer	medical@ncpor.in	Med@2025	Personnel, Safety, Emergency
 Operator	operator@ncpor.in	Op@2025	Dashboard, Alerts, Inventory

KEY DEMO HIGHLIGHTS

- Login with any role → see role-based dashboard
 - Watch **live sensor tiles** update every 5 seconds
 - Click **Digital Twin map** buildings to see live data
 - Trigger **SOS / emergency protocols** in Emergency module
 - Run a **What-If simulation** (e.g., Generator Failure)
 - Add a **logbook entry** with Ctrl+Enter shortcut
- Check **AI Predictive Maintenance** risk table in Equipment
 - View **30-day fuel forecast** chart in Inventory
 - Acknowledge and resolve **active alerts**
 - Test **report generation** modal in Reports
 - Toggle **Collapse** sidebar for more screen space
 - View **anomaly detection** tab in Environmental

PROJECT SUMMARY STATISTICS

~1,500

LINES OF CODE

Single HTML file

0

CDN DEPENDENCIES

100% offline ready

5s

UPDATE INTERVAL

Live sensor refresh

288

HISTORY POINTS

Per sensor (24h)

ACKNOWLEDGEMENTS

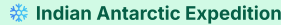
We gratefully acknowledge **NCPOR (National Centre for Polar & Ocean Research)**, Goa — an autonomous institution under the **Ministry of Earth Sciences, Government of India** — for their pioneering work in Antarctic research and for inspiring this project. Special thanks to the Indian Antarctic Expedition teams at Maitri and Bharati stations whose operational challenges guided our design decisions.

 NCPOR

 MoES

 IN Govt. of India

 SIH 2025

 Indian Antarctic Expedition